Roll Number: _____

Thapar Institute of Engineering and Technology (TIET), Patiala
Department of Computer Science & Engineering (CSED)
AUXILIARY EXAMINATION

B.Tech Auxiliary Examination (2024-25)	**Course Code**: UCS701 **Course Name**: Theory of Computation
February 21, 2025	Friday, 05:30 PM-08:30 PM
Time: 3Hrs, Max. Marks: 100	**Name of Faculty**: Dr. Shashank Sheshar Singh

Note: All questions are compulsory. All parts of a question should be attempted in one place. Assume missing data, if any suitably.

Q. 1 Define deterministic finite automata. Construct a DFA over $\sum = \{a,b\}$ where every a is followed by bb. **[10]**

Q. 2 Define push-down automata. Construct a push-down automata for a language L= $\{0^n 1^m \mid \Sigma = \{0, 1\}\}$. **[10]**

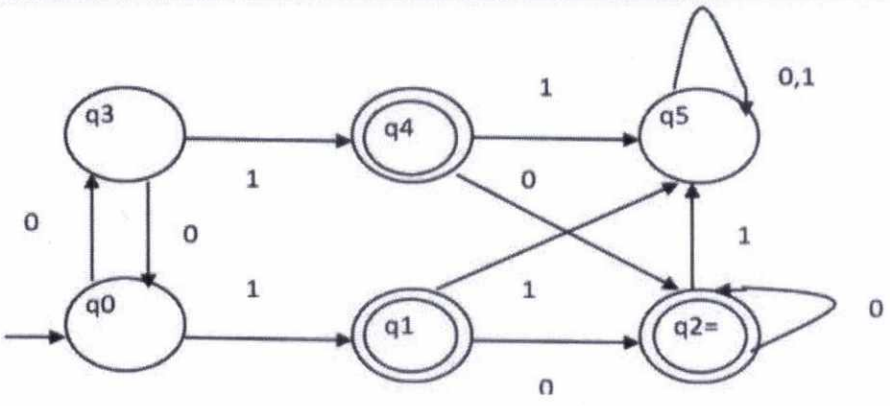
Q. 3 Define Turing machine. Construct a Turing machine for a language L= $\{0^n 1^n 0^n \mid \Sigma = \{0, 1\}\}$. **[10]**

Q. 4 Consider following context-free grammar to CNF. **[10]**

$$A \rightarrow BAB \mid B \mid \varepsilon$$
$$B \rightarrow 00 \mid \varepsilon$$

Q. 5 Construct null-NFA using the Thompson construction method for L= (a+b)*(ab)*. Also find DFA corresponding to null-NFA generated by Thompson construction method. **[10]**

Q. 6 Find equivalent minimal DFA for following finite automata. **[10]**



Q. 7 Construct a Moore machine of residues of 3 of a given binary number. **[10]**

Q. 8 Define the pumping lemma for regular language. Show that the language L = {ww|w ϵ {0,1}*} is not a regular language using the pumping lemma. **[10]**

Q. 9 Draw Finite Automata for languages: **[10]**
1. L1={x|00 is not substring of x, x ∈ {0,1}*}
2. L2={x| x ends with 01, x ∈ {0,1}*}

Also, draw FA for $L1 \cup L2$ and $L1 - L2$. **[10]**